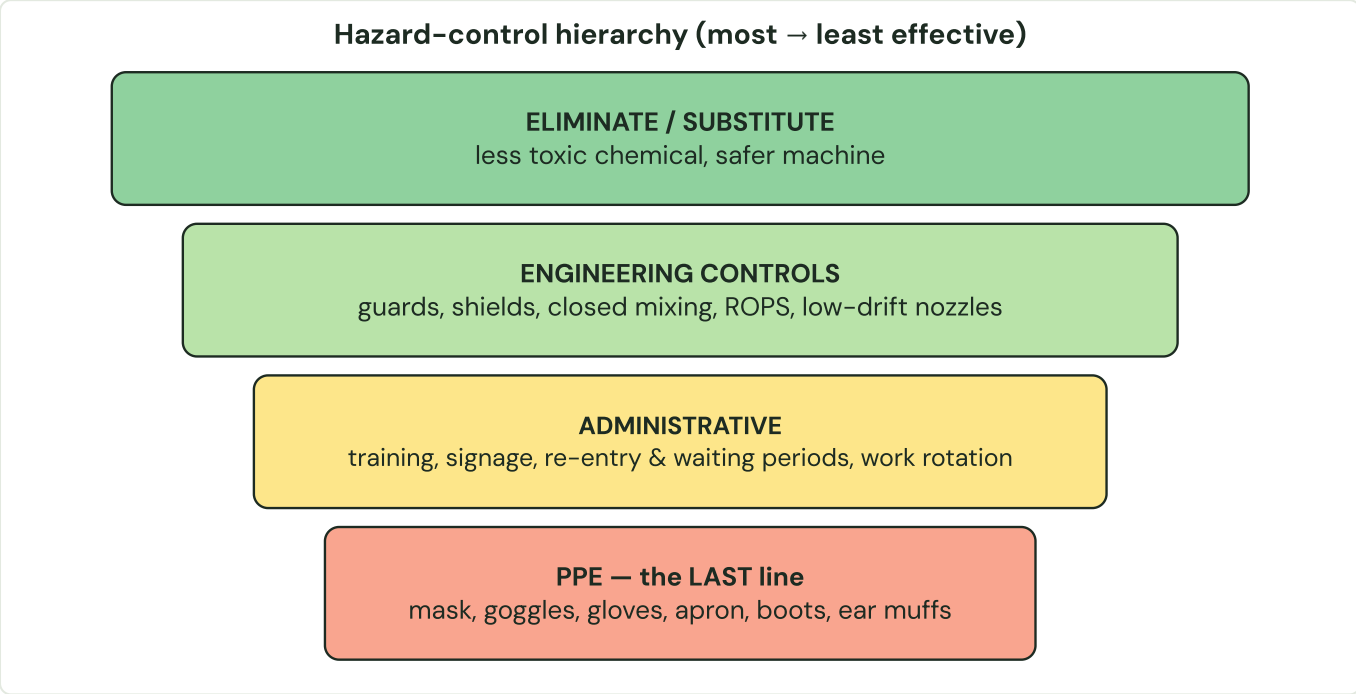
UNIT 5 · TOPIC 8

Operator's Protective Gadgets (PPE)

Syllabus topic: Operator's protective gadgets



Hazard-control hierarchy — PPE is the last line of defence (redrawn from the source figure)

Kit for spraying

- Respirator / cartridge mask — the correct cartridge for the chemical.
- Goggles or face shield.
- **NITRILE or rubber gloves** — cotton gloves ABSORB the chemical and worsen exposure.
- Full-sleeve overall/apron and hood; gum boots.

Kit for machinery work

- Ear muffs/plugs above ~**90 dB(A)** (Topic 6a).
- Safety shoes; close-fitting clothing (no loose ends near shafts).
- Gloves for handling blades and sheet metal.
- Dust mask in threshing and winnowing.
- On the tractor: seat belt with ROPS; SMV emblem and lights (reflective jacket) for road travel.
- Broad-brim hat for sun protection; first-aid kit and clean water on site.

Limitations of PPE (exam angle)

- PPE is the LAST resort. It fails if uncomfortable in Indian heat, ill-fitting, poorly maintained or unavailable.
- Hence CIAE-type designs stress lighter, breathable, affordable protective clothing — and, better still, engineering controls that need no operator discipline.

Practice MCQs

Q103. Which gloves are correct for pesticide handling?

(a) Cotton
(b) Nitrile/rubber
(c) Woollen
(d) None

Q105. In the hazard-control hierarchy, PPE ranks:

(a) First
(b) Second
(c) Third
(d) Last — after elimination, engineering and administrative controls

Answers

Q103 — (b) Nitrile/rubber — *Cotton absorbs and holds the chemical against the skin.*

Q105 — (d) Last — after elimination, engineering and administrative controls

Sources: Liljedahl et al. — Tractors and their Power Units, Ch. 9, pp. 231–235; ASAE S383.1 & S310.3; BIS safety standards IS 9020 / IS 9129 / IS 12239; Insecticides Act 1968 & Rules 1971 (label classes); WHO hazard classification; ICAR-CIAE / AICRP farm-accident surveys; NPTEL lectures.